

Problem Set 7

E. Evacuation

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tianzheyu submissions

#	When	Who	Problem	Lang	Verdict	Time	Memory
383789178	Jul/22/2026 17:04 ^{UTC+10}	tianzheyu	E - Evacuation	C++23 (GCC 14-64, msys2)	Accepted	46 ms	100 KB

<https://codeforces.com/submissions/tianzheyu#>

Process

Implemented a time expanded network utilizing Dinic's algorithm to model dynamic movement over time, combined with a preliminary BFS to calculate spatial constraints (gas spreading times).

Challenges and Overcoming

- When mapping my multi-dimensional states to 1D node IDs, I ran into a bug where my graph's edges were being improperly overwritten and misdirected. I found it when I reviewed my node index allocations and realized I used $n \times n \times t$ as the base offset for my super source, super sink, and rescue capsule nodes. This directly collided with the grid nodes at the final time step t (since time spans from 0 to t , it requires $t+1$ layers). Next time to prevent this bug, I will explicitly calculate the maximum possible size of the state space (e.g., $(t+1) \times n \times n$) and define a constant base offset strictly above this upper bound before assigning any special node IDs.
- Later, when defining the valid movement edges, I ran into a bug where my program outputted a lower survivor count than expected on edge cases. I found it when I traced the timing of the gas spread and realized my movement condition `k < visited[target]` strictly prevented a scientist from reaching a capsule at the exact minute the gas arrived, even though the problem logic permits escaping in that final moment before infection. Next time to prevent this bug, I will manually diagram the exact boundary conditions of simultaneous events on a timeline to ensure my code's inequalities (`<` versus `<=`) perfectly reflect the problem's physical rules.